

Linux Commands Cheat Sheet with Examples & Uses

1. Basic Commands

- pwd - Print Working Directory

Example: pwd

Use: Shows your current location in the filesystem.

- ls - List Directory Contents

Example: ls -l /var/log

Use: Lists files and folders with permissions, size, and date.

- cd - Change Directory

Example: cd /etc/nginx

Use: Moves between directories.

- mkdir - Create Directory

Example: mkdir /opt/newfolder

Use: Creates a new folder.

- touch - Create File

Example: touch newfile.txt

Use: Creates an empty file.

- cat - View File Content

Example: cat /etc/passwd

Use: Displays file contents in terminal.

2. File Management

- cp - Copy Files or Directories

Example: cp file.txt /backup/

Use: Copies files to another directory.

- mv - Move or Rename Files

Example: mv oldname.txt newname.txt

Use: Moves or renames files.

- rm - Remove Files or Directories

Example: rm -ri /tmp/test

Use: Deletes files; '-r' for recursive, '-i' for interactive.

- find - Search Files

Example: `find / -name nginx.conf`

Use: Searches for files anywhere in the filesystem.

- du - Show Directory Size

Example: `du -sh /var/log`

Use: Displays disk usage of directories.

- df - Show Disk Space

Example: `df -h`

Use: Displays total, used, and free space on mounted filesystems.

3. User & Permission Management

- useradd - Add a New User

Example: `sudo useradd devuser`

Use: Creates a new user account.

- passwd - Change Password

Example: `sudo passwd devuser`

Use: Changes or sets a user's password.

- chown - Change Ownership

Example: `sudo chown tomcat:tomcat /opt/tomcat -R`

Use: Changes file/directory ownership.

- chmod - Change Permissions

Example: `chmod 755 script.sh`

Use: Sets permissions for files or directories.

- groups - List User Groups

Example: `groups devuser`

Use: Displays groups a user belongs to.

4. Process & System Monitoring

- ps aux - Show Running Processes

Example: `ps aux | grep nginx`

Use: Lists all running processes with details.

- top - Real-time System Monitor

Example: `top`

Use: Displays CPU/memory usage by processes live.

- kill - Terminate Process

Example: kill 1234

Use: Stops a process using its PID.

- free - Show Memory Usage

Example: free -h

Use: Displays used, free, and total memory.

- uptime - Show System Uptime

Example: uptime

Use: Shows how long the system has been running.

5. Networking

- ping - Test Connectivity

Example: ping google.com

Use: Checks if a host is reachable.

- ip a - Show Network Interfaces

Example: ip a

Use: Displays IP addresses and network adapters.

- ss - Show Open Ports

Example: ss -tulnp

Use: Displays listening ports and associated services.

- curl - Fetch Data from a URL

Example: curl -I https://example.com

Use: Tests web server or API connectivity.

- scp - Secure File Copy

Example: scp file.txt user@192.168.1.10:/tmp

Use: Transfers files securely between systems.

- ssh - Remote Login

Example: ssh user@server_ip

Use: Connects securely to remote servers.

6. Service Management

- systemctl start <service>

Example: `sudo systemctl start nginx`

Use: Starts a service.

- `systemctl stop <service>`

Example: `sudo systemctl stop nginx`

Use: Stops a running service.

- `systemctl restart <service>`

Example: `sudo systemctl restart httpd`

Use: Restarts a service.

- `systemctl enable <service>`

Example: `sudo systemctl enable nginx`

Use: Enables service to start at boot.

- `systemctl status <service>`

Example: `sudo systemctl status tomcat`

Use: Shows service current state.

7. Compression & Archiving

- `tar` - Archive Files

Example: `tar -cvf backup.tar /var/www`

Use: Creates a tar archive.

- `tar` - Extract Files

Example: `tar -xvf backup.tar`

Use: Extracts files from a tar archive.

- `gzip` - Compress File

Example: `gzip file.txt`

Use: Compresses file to file.txt.gz.

- `gunzip` - Decompress File

Example: `gunzip file.txt.gz`

Use: Decompresses a .gz file.

- `zip/unzip` - Compress/Extract ZIP Files

Example: `zip -r backup.zip /home/user`

Use: Creates zip archives; unzip extracts them.